S05. Finite-basis density functionals and exact convex constraints

Pure Thought research portfolio

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Source: synthesis challenge 17; original synthesis. This topic is distinct from, or adds a distinct objective to, the canonical PRD catalog. Crosswalk

Plain-language guide

The problem in everyday terms

A many-electron state contains vastly more information than the average number of electrons at each site. Can we still describe the minimum energy using only those average occupancies? This problem studies that question in a finite lattice, where the exact relationship can be defined and bounded without pretending that a convenient approximate formula is exact.

Key terms

- **Density** — Here, the expected electron occupancy at each lattice site.
- **Density functional** — A rule assigning an energy contribution to a full density profile.
- **Finite basis** — A fixed finite collection of states or orbitals used to define the model.
- **Ensemble** — A statistical mixture of allowed quantum states.
- **Constrained search** — Minimization over all states with a prescribed density.
- **Kinetic energy** — The energy associated with electron motion or hopping in the model.
- **Interaction energy** — The energy arising from interactions between electrons.
- **Convexity** — The property that mixing two density profiles cannot raise the functional above the corresponding mixture of functional values.
- **Duality** — An equivalent optimization viewpoint using external potentials to constrain energy values.
- **Exchange–correlation energy** — A remainder defined only after specifying the noninteracting kinetic and Hartree reference contributions.

Why this matters

Density-based methods reduce a difficult many-electron description to a much smaller set of variables. Exact finite examples reveal which constraints an approximation should respect and provide benchmarks for evaluating proposed energy formulas.

What progress would mean

A useful result would bracket the functional on the declared density domain and verify its convex and dual structure. It would give a controlled model benchmark, with a separate argument needed before extending conclusions to continuum chemistry or real materials.

Revised core question

For a fixed finite lattice Hamiltonian, can the ensemble constrained-search density functional be bracketed and its convexity/duality verified?

Precise scope, target and outputs

Fix a finite N-particle Hilbert space, kinetic T, interaction W, and commuting site-density operators $\hat{n}_i$. Define $F(n) = \min_{\rho\succeq 0, \mathrm{Tr}\,\rho=1, \mathrm{Tr}(\rho\hat{n}_i)=n_i} \mathrm{Tr}\,\rho(T+W)$ on ensemble-representable densities. Derive the dual $F(n) = \sup_v[E(v) - v\cdot n]$ with $E(v) = \min_{\rho\succeq 0, \mathrm{Tr}\,\rho=1} \mathrm{Tr}\,\rho(T+W+\sum_i v_i\hat{n}_i)$, under the finite-dimensional convex formulation. Use exact states for primal upper bounds and potentials with certified E(v) lower bounds for dual lower bounds. If defining an exchange-correlation remainder, specify T_s and Hartree conventions separately.

Detailed critique of the synthesis framing

"The cone of exchange-correlation functionals satisfying all exact constraints" is not justified: normalization and affine conditions need not define a cone, and convexity of the ensemble universal functional does not imply convexity of E_xc after subtracting other functionals. Constraint satisfaction alone does not identify the exact physical functional. The broad continuum task also leaves admissible density spaces and spin scaling unspecified. The revision first tests exact convex duality in a finite model, preserving nonempirical functional development as an extension.

Validation and milestones

Baseline: exact functional values and dual brackets on a two-site model. Strong: a certified piecewise characterization or a new valid inequality for a larger finite model. Verify particle-number bounds, convexity inequalities, and agreement with direct constrained search.

Use the shared evidence standard. Unsuccessful search is unresolved unless a complete exclusion proof is supplied. These are proposed baseline specifications, not completed scientific results.

Displayed formulation and claim boundary

Supplemental specification.

$$F(n) = \min_{\substack{\rho\succeq 0,\ \mathrm{Tr}\,\rho=1 \\ \mathrm{Tr}(\rho\hat{n}_i)=n_i}} \mathrm{Tr}\,\rho(T+W),$$

$$E(v) = \min_{\rho\succeq 0,\ \mathrm{Tr}\,\rho=1} \mathrm{Tr}\,\rho\left(T+W+\sum_i v_i\hat{n}_i\right), \qquad F(n) = \sup_v[E(v) - v\cdot n].$$

Work in the fixed finite N-particle Hilbert space and on ensemble-representable densities. The supremum need not be attained by a finite potential at boundary densities. Feasible density matrices give upper bounds; certified energy lower bounds at chosen potentials give lower bounds. Convexity of F does not imply convexity of an exchange-correlation remainder.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means "not excluded"; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in

this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.